



Learning Objectives

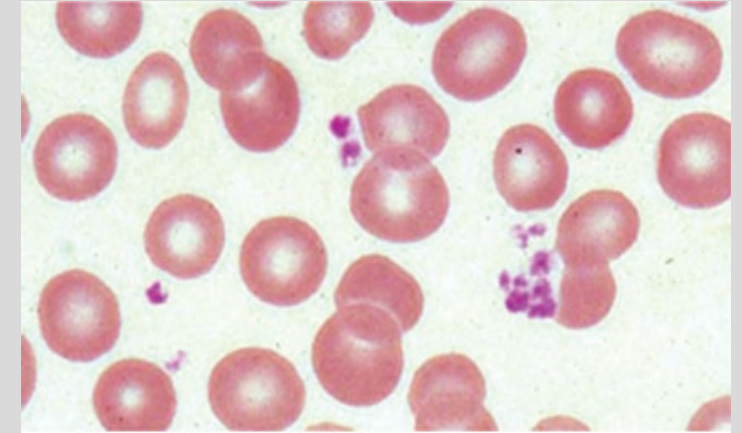
1. Identify the most common causes of microcytic, normocytic, and macrocytic anemia.
2. Describe the pathophysiology, clinical symptomatology, and laboratory findings of the various causes of microcytic, normocytic, and macrocytic anemia.
3. Illustrate the biochemical basis of heme synthesis, vitamin B12 and folate deficiency, and the clinical approach to such patients.



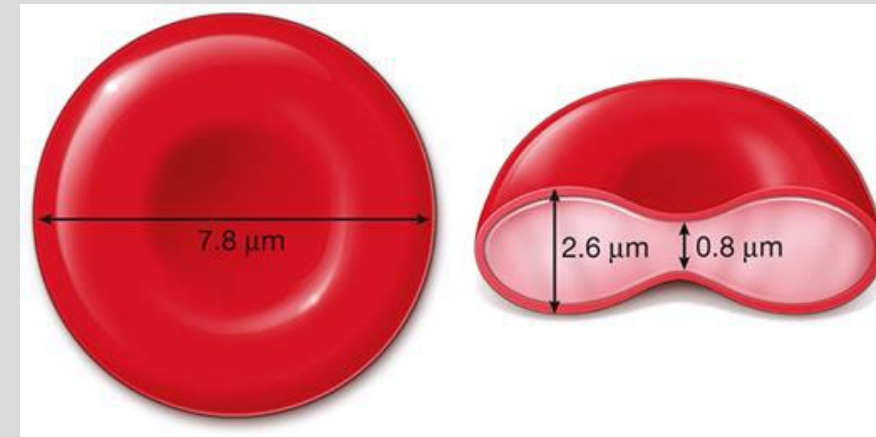
Erythrocytes (Red Blood Cells)

- **Function:** deliver O_2 from the lungs to peripheral tissues and CO_2 from the peripheral tissues to the lungs
- **Biconcave shape**, anucleate cells and lack organelles.
 - This is to increase the amount of cytoplasm in the cell
- They are flexible, important when navigating small capillaries.
 - This is due to special membrane proteins (more to follow)
- Life Span: ~ 120 Days
 - In healthy individuals, about ~1% of all RBCs are removed each day due to senescence (they're old!)

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Source: Jon C. Aster, H. Franklin Bunn:
Pathophysiology of Blood Disorders, Second Edition
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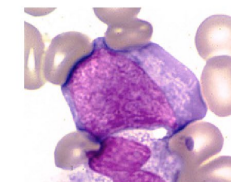
Erythrocyte Life Cycle



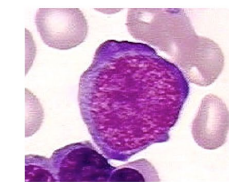
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- RBCs are produced in the bone marrow (erythropoiesis)
- Erythropoietin (**EPO**) growth factor helps stimulate production
 - Decreased O_2 is sensed by interstitial cells in the peritubular capillary bed of the **renal cortex** → EPO secreted
 - EPO & hematocrit levels are inversely proportional
- **Reticulocytes (immature RBCs)** are released into circulation
 - In ~24-48 hours they mature to erythrocytes
 - Reticulocyte count is **1-2%** of total RBCs
 - Reticulocytes appear more basophilic (purple) compared to mature RBCs **due to rRNA still present**
- Senescent (old) RBCs are removed by macrophages in the spleen, liver, and bone marrow
 - Reticuloendothelial System (RES) recycles the RBCs

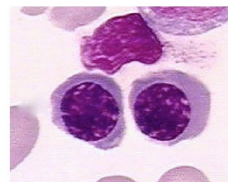
	Normoblast	Reticulocyte	Mature RBC
			
Nuclear DNA	Yes	No	No
RNA in cytoplasm	Yes	Yes	No
In marrow	Yes	Yes	Yes
In blood	No	Yes	Yes



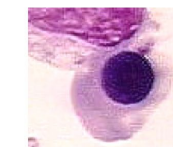
Proerythroblast



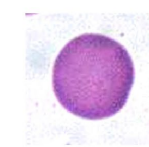
Basophilic erythroblast



Polychromatophilic erythroblast



Orthochromatophilic erythroblast



Reticulocyte



Erythrocyte



RBC Indices

Size of RBCs:

1. **Mean Cell Volume (MCV)**: average **size** of RBCs
 - Both Genders: 80-100 fL
2. **Red Blood Cell Distribution Width (RDW)**: measures the **variability** of the MCV
 - Both Genders: 12-15%
 - Shows if the erythrocytes are all the same or if they are different in sizes or shapes
 - **Higher RDW** means large range of size

Amount of hemoglobin per RBC:

1. **Mean Corpuscular Hemoglobin (MCH)**: amount of Hgb in an average erythrocyte
2. **Mean Corpuscular Hemoglobin Concentration (MCHC)**: provides the percentage of Hgb concentration in an average erythrocyte

Other:

1. **Reticulocyte Count**: the percentage of total RBCs that are reticulocytes (immature RBCs)
 - This is important when **determining if the bone marrow is responding appropriately** to anemia
 - Anemia with ↑ reticulocyte count, bone marrow is **responding appropriately**
 - Anemia without ↑ reticulocyte count, bone marrow may be the **issue** for the anemia
 - Remember, bone marrow takes time to respond, so there will be a **DELAY**



Anemia Classification

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- When working someone up for potential anemia, the first test you will order is a CBC
- If the hemoglobin/hematocrit/RBC is **LOW**, **your patient is anemic by definition**
- To determine the type of anemia your patient may have, the next best lab value to examine is the **mean cell volume (MCV)**
 - The MCV measures the average size of the RBCs



Microcytic = MCV <80

Normocytic = MCV 80-100

Macrocytic = MCV >100





Microcytic Anemias

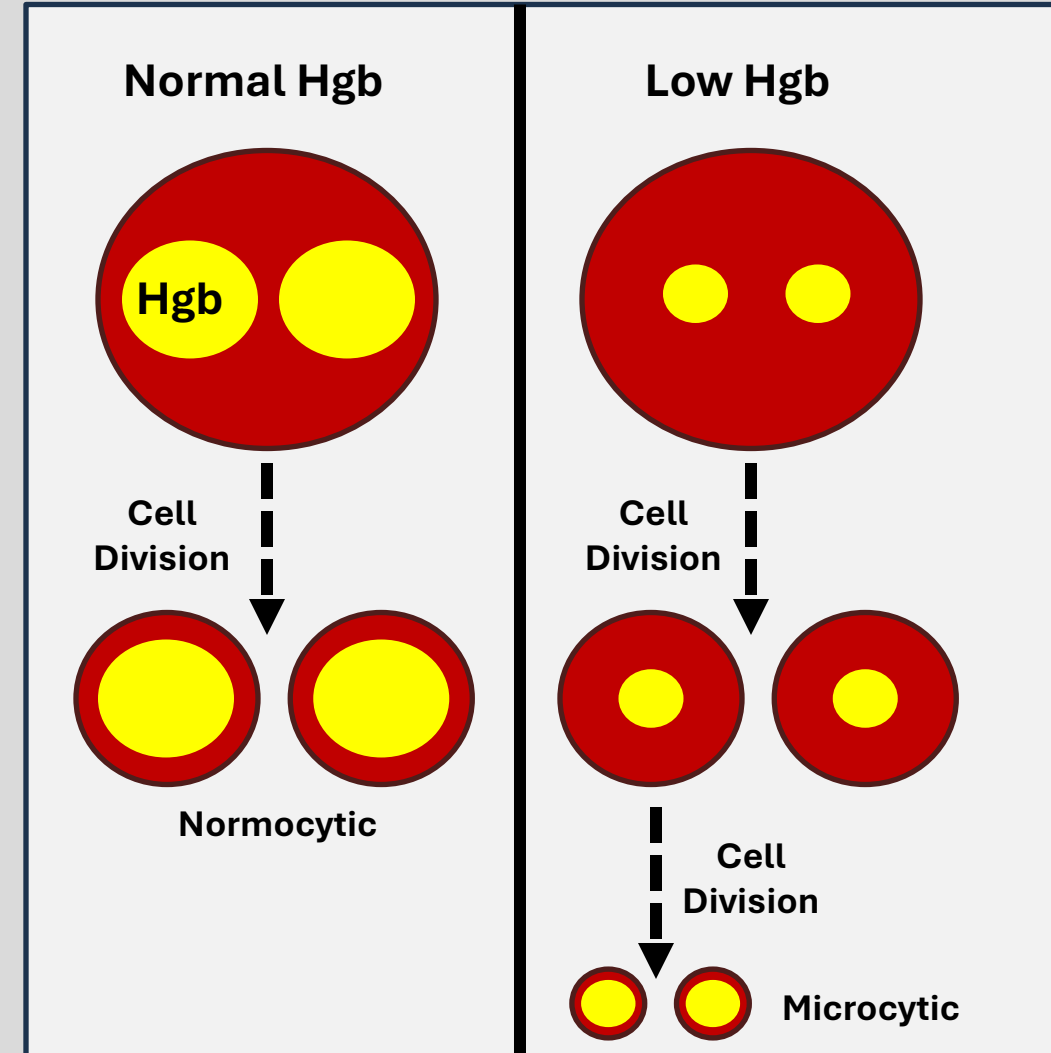
1. Iron Deficiency Anemia (Later Stages)
 2. Anemia of Chronic Disease (Later Stages)
 3. Lead Poisoning
 4. Sideroblastic Anemia
 5. Thalassemia
-
- Your red blood cells are **smaller than normal!**
 - For anemia of chronic disease and iron deficiency anemia, you will typically see a **higher RDW, as these two anemias start as normocytic**
 - Normocytic anemia has larger RBCs, as the disease progresses the RBCs become smaller
 - Therefore, we have a greater mix (higher variance) of RBC size in the blood, increasing the RDW



RBC Division

- **Erythroblasts:** RBC progenitor cells in the bone marrow
 - As the RBC divides and matures, they become smaller
 - This is due to RBCs maintaining a **specific concentration of hemoglobin** in the cell to function properly
- **Microcytic Anemia:**
 - Due to the **decreased production of hemoglobin**
 - Since there is less hemoglobin, the RBC must undergo additional cell division to maintain the **concentration of hemoglobin** in the cell

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Due to lower hemoglobin concentration, additional cell division necessary



Labs for Microcytic Anemia

- Once a diagnosis of microcytic anemia is made, the next best step is to order an **iron study**

Results of an Iron Study:

- Serum Iron:** measures the amount of iron in the blood
- Serum Ferritin:** this reflects the amount of iron stored in the bone marrow macrophages and liver
 - Remember, ferritin is the protein that **stores iron**
- Total Iron Binding Capacity:** the amount of transferrin circulating in the blood
 - TIBC is transferrin**
 - Transferrin is made in the liver. There is a relationship between iron storage (ferritin) and TIBC. When iron stores in the bone marrow are low, signals are sent to the liver to produce **MORE** transferrin. Therefore, transferrin is increased (hence why TIBC is **high** in iron deficiency anemia)
- % Saturation:** the percentage of transferrin that is bound to iron (~33% is normal)
 - This is a calculation ($\frac{\text{serum iron}}{\text{TIBC}}$)
- Ferritin and TIBC are generally opposite**
 - If we have **increased ferritin** (increased iron **storage**), our **binding capacity (TIBC)** is **low**
 - Liver will produce less transferrin because cells are full of iron)
 - If we have **decreased ferritin** (decreased iron **storage**), our **binding capacity (TIBC)** is **high**
 - Liver gets signals to make more transferrin because cells need more iron

Iron Deficiency Anemia

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- Without iron, the body cannot produce **heme**
 - **Most common cause of anemia worldwide** (>12% of world's population*)
- **Iron loss > Iron intake**
- **Iron excretion is not regulated**
 - Bleeding (GI bleeding/cancer; menses in females)
 - Loss of epithelial cells from skin, gut, genitourinary tract
- **Can be due to iron intake issues**
 - Iron is absorbed in the **duodenum**
 - **Any issue with the duodenum can affect uptake** (disease, surgery)
 - **Inadequate nutritional iron intake**
- **General Treatment: oral iron supplement**

**LITTLE MISS
ANEMIC**



Iron Metabolism

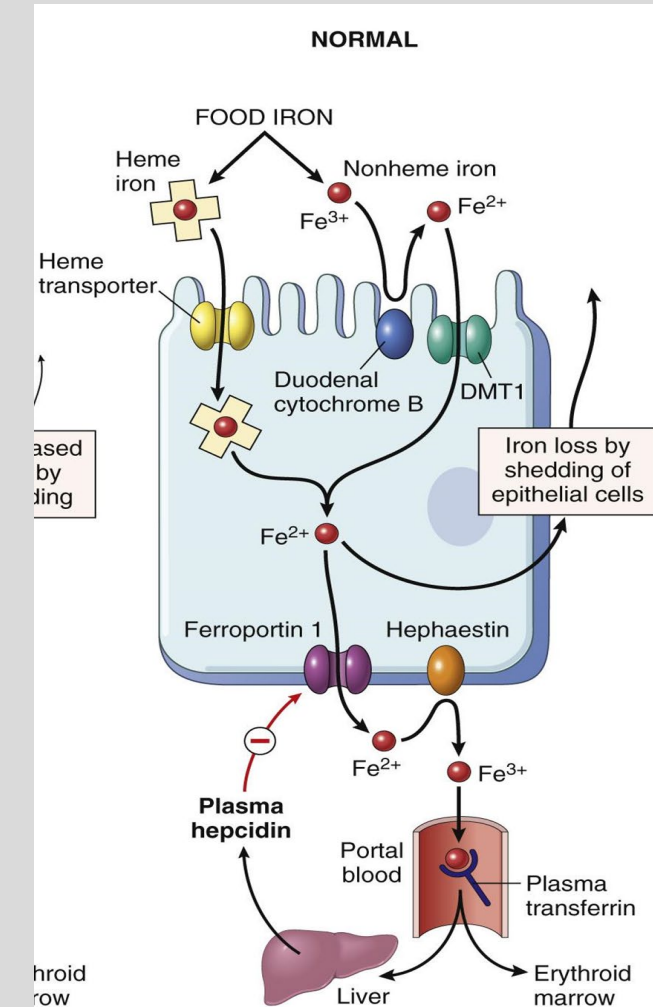
Iron: Iron absorbed in **duodenum**

Fist: **Folate** (B_9) absorbed in **small bowel**

Bro: B_{12} absorbed in terminal **ileum**

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- Iron in the body is recycled between circulated and stored pools.
- Iron is carried by **transferrin** in the plasma. Transferrin is a protein synthesized in the liver
 - Main **function** of transferrin: **deliver iron to cells** to help synthesize hemoglobin
- Iron Transport from the GI System
 1. Iron is consumed in heme (meat) and non-heme (vegetable)
 2. Iron is absorbed by enterocytes in the **duodenum**
 3. **Non-heme iron (Fe^{3+})** must be reduced from Fe^{3+} to Fe^{2+} by ferrireductase (cytochrome B) on the surface of the enterocyte. It is then **transported into the cell by DMT1**
 4. **Heme** is absorbed by transporters into the enterocyte
 5. All iron leaves the enterocyte via **ferroportin**
 6. Once in the plasma, iron is transported via **transferrin**
 7. **Transferrin** will take iron to its final destination (macrophages in the marrow and liver), **iron will be stored by ferritin**
- **Hepcidin:** **regulates this process** by decreasing ferroportin



Usual Causes of Iron Deficiency

Major Causes:

- **Blood loss**
- **Dietary deficiency** (most iron comes from meat and vegetables)

Most Common Causes by Age

- **Infants:** breast feeding (human milk) is naturally low in iron
- **Children:** poor diet
- **Adults (20-50):**
 - Males: peptic ulcer disease
 - Females: menorrhagia (increase blood loss w/ menses) or pregnancy (increase overall iron demands)
- **Elderly:**
 - Western world: colon polyps/cancer
 - Developing world: hookworms
- **Clinical Pearl: any elderly individual with iron deficiency anemia is deemed to have colon cancer until proven otherwise*

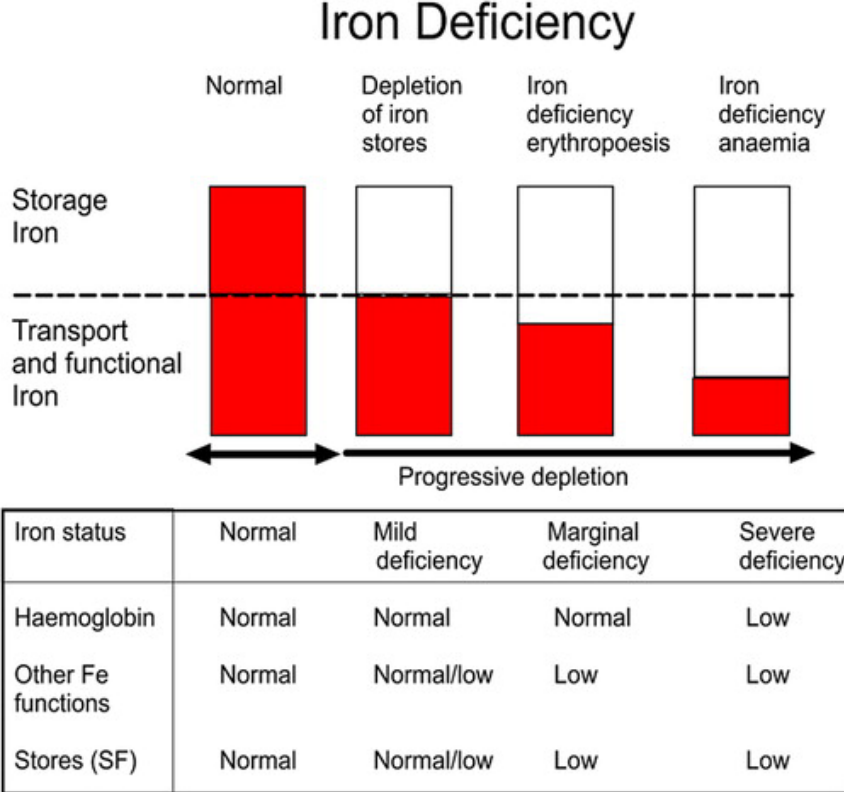
Other Causes to Keep in Mind:

- Malnutrition
- Malabsorption → celiac disease
- Surgery → gastric bypass surgery (food bypasses duodenum) and gastrectomy (stomach acid helps increase Fe^{2+})



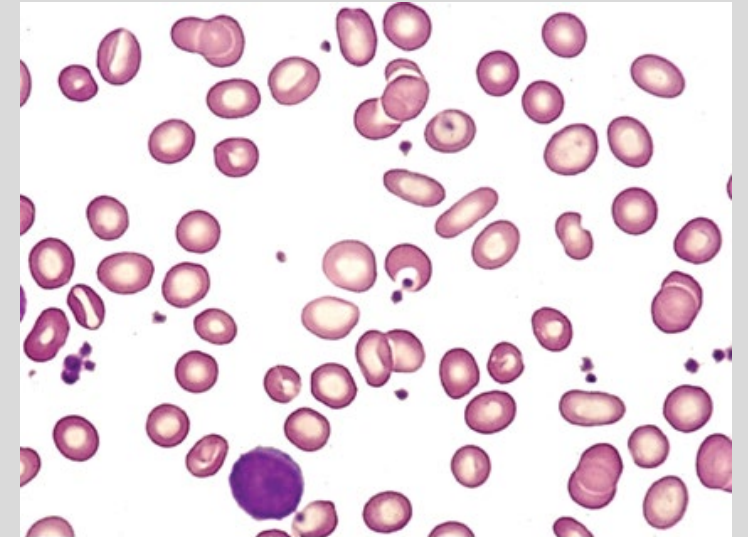
Stages of Iron Deficiency

- 1. Storage of iron is depleted in the cells
 - Ferritin ↓
 - TIBC ↑
 - Cells will recruit more iron from the plasma
- 2. Serum iron is depleted
 - Serum iron ↓
 - % Saturation ↓
 - TIBC ↑
- 3. Normocytic Anemia (Early Stage): the body will begin to make normal sized RBCs, but will reduce the quantity.
 - MCV: 80-100
 - Hemoglobin/Hematocrit/RBC: ↓
 - RDW: normal
- 4. Microcytic/Hypochromic Anemia (Later Stage): bone marrow will make smaller RBCs to conserve the concentration on hemoglobin, and fewer cells are made
 - **MCV: <80**
 - **RDW: ↑**
 - Remember, RDW examines the variation in size. Since RBCs live for ~120 days, we will have a mix of normocytic cells and microcytic cells, increasing the variations in size



Laboratory Findings of IDA

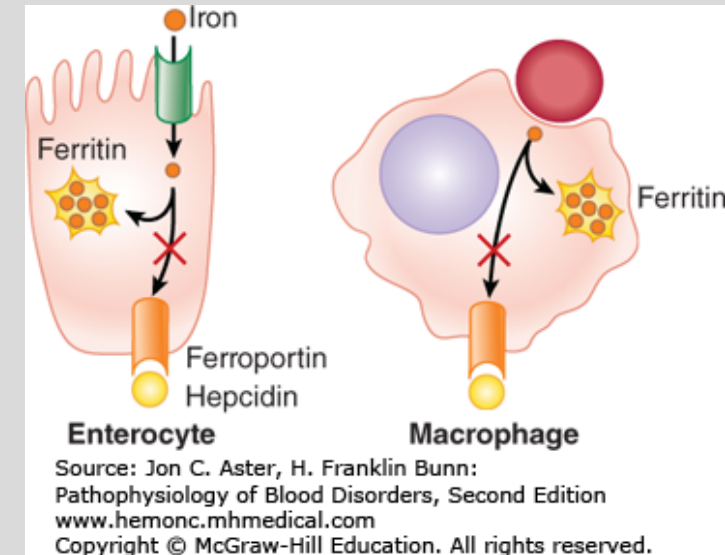
- **Microcytic Anemia (MCV <80) in the later stages**
- **Peripheral blood smear** will show **microcytic** and **hypochromic** RBCs
 - Less pigmentation due to less hemoglobin
- **Ferritin ↓**
 - We have depleted our stores of iron
- **TIBC ↑**
 - The liver is producing more transferrin in an attempt to shuttle more iron to cells
- **% Saturation of Transferrin ↓**
 - Less iron binding to transferrin due to overall iron deficiency
- **Free Erythrocyte Protoporphyrin ↑**
 - Protoporphyrin combines with iron to create heme
 - Less iron will stop this reaction, increasing the amount of reactants (protoporphyrin)
- **RDW ↑**
 - Increased due to the mixed sizing of the RBCs
 - The marrow produced normocytic cells earlier on in the anemia, now shifting to microcytic. Mix of sizes, increasing variance.



Source: Jon C. Aster, H. Franklin Bunn:
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Anemia of Chronic Disease

- Anemia associated with chronic inflammation or disease. Most common anemia in hospitalized patient
- **Pathophysiology:** inflammatory mediators (**IL-6**) released during inflammation/infection limit free iron via increased hepcidin production
 - Hepcidin sequesters iron in the storage cells via
 1. Limiting iron transfer from macrophages to erythroid precursor cells
 2. Suppressing EPO production
 - This response to inflammation is an evolutionary trait, as bacteria need iron for survival. Immune system wants to limit bacterial growth
 - Iron is locked away in the macrophages of the bone marrow
- **Labs:**
 - **TIBC ↓**: liver will produce less transferrin because our stores are high
 - **Ferritin ↑**: holding iron in the bone marrow
 - **% Saturation ↓**: less iron in circulation
 - **Serum Iron ↓**: iron is sequestered in the bone
- **Treatment:** treat underlying disease/inflammation



Anemia of Chronic Disease vs Iron Deficiency

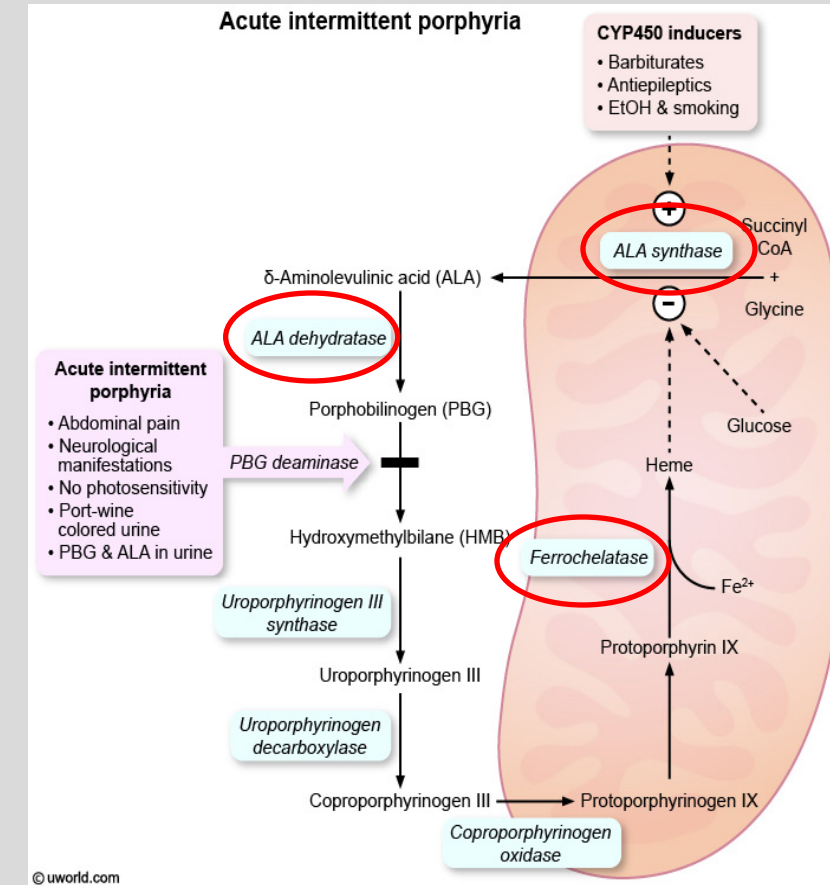
- Although lab values are different, the main issue is similar: there is a lack of available iron to be used for hemoglobin synthesis
- Key differentiators:
 - Can be via clinical vignette, thinking of iron deficient anemia in patients with risk factors. Think of anemia of chronic disease in patients with long term disease, kidney disease
 - **Lab Values are Key**
 - **Ferritin** is **LOW** in IDA and **HIGH** in ACD
 - **TIBC** is **HIGH** in IDA and **LOW** in ACD
 - ***Don't just memorize lab values, think through why each value moves the way it does***

State	Serum Iron	Ferritin	TIBC	% Saturation
Normal	100 $\mu\text{g}/\text{dL}$	-	300 $\mu\text{g}/\text{dL}$	33%
Iron Deficient Anemia	↓	↓	↑	↓
Anemia of Chronic Disease	↓	↑	↓	↓

Sideroblastic Anemia

- Pathophysiology: defective protoporphyrin synthesis
 - The issue still relies in making heme
- **Causes:**
 - **Congenital defect:** most common is defect in ALAS (X-linked)
 - **Acquired Causes:**
 - **Alcoholism:** poisons the mitochondria
 - **Lead:** inhibits ALAD and ferrochelatase
 - **Vit B6 Deficiency:** needed for ALAS
 - Commonly seen as a side effect of tuberculosis treatment (isoniazid)

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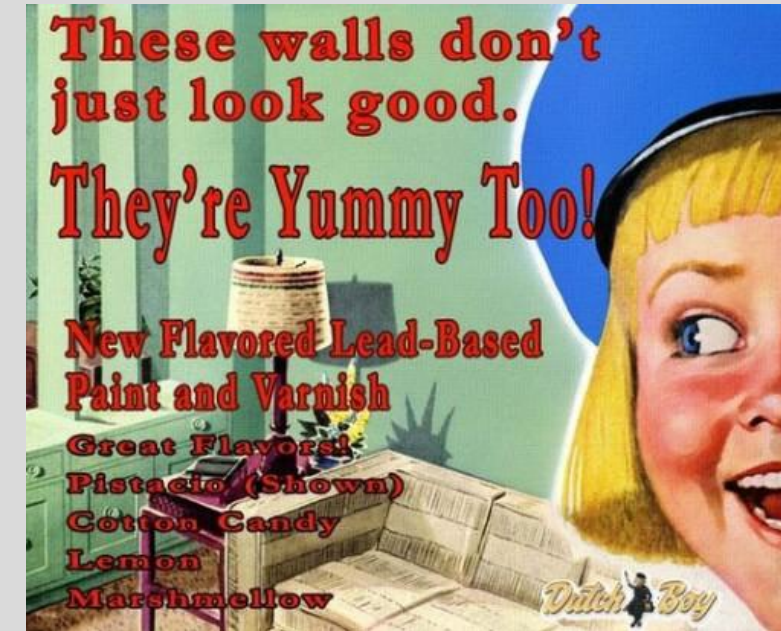




Lead Poisoning

- Lead inhibits **ferrochetalase** (last step) and **ALAD** (second step)
 - ↓ heme synthesis
 - ↑ RBC protoporphyrin: the protoporphyrin is made, but cannot be attached to iron in the mitochondria
 - Inhibits rRNA degradation: causes basophilic stippling
- **Symptoms (LLEEAAD):**
 - Lead lines on gingivae and metaphysis of long bones
 - Encephalopathy (confusion) and erythrocyte basophilic stippling
 - Abdominal colic and sideroblastic anemia
 - Drops: wrist and foot drop (neuropathies)
- Exposure: children in older houses (built before 1978) with chipped paint and occupation in adults

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Lead line in the gums of a patient with occupational lead poisoning



Photograph provided by Stephen A Landaw, MD PhD UpToDate®

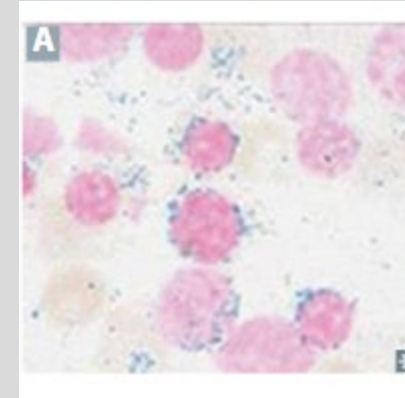
Sideroblastic Anemia Lab Findings

- MCV <80
- **TIBC ↓**: liver will produce less transferrin because our iron stores are high
- **Ferritin ↑**: there is excess iron, the bone will attempt to store more
- **% Saturation ↑** : generally, there is iron overload
- **Serum Iron ↑** : generally, there is iron overload
 - Remember, our issue lies in protoporphyrin production. The body has enough iron to create heme
 - IRON OVERLOAD!

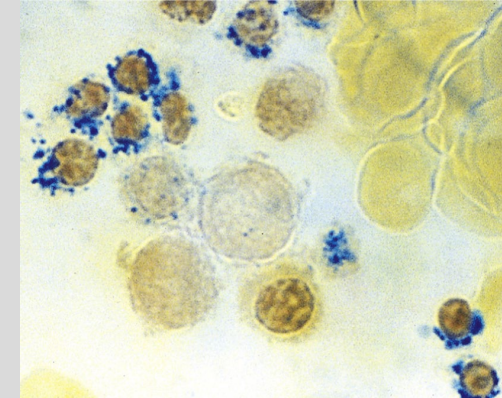
State	Serum Iron	Ferritin	TIBC	% Saturation
Normal	100 $\mu g/dL$	-	300 $\mu g/dL$	33%
Iron Deficient Anemia	↓	↓	↑	↓
Anemia of Chronic Disease	↓	↑	↓	↓
Sideroblastic Anemia	↑	↑	↓	↑

Sideroblastic Anemia Findings

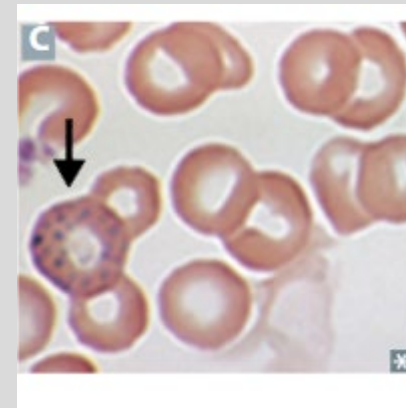
- **Bone Marrow Aspirate:** ringed sideroblasts appear with **Prussian Blue Stain**
 - Due to excess iron accumulating in perinuclear mitochondria
 - **Need Prussian blue stain**
- **Peripheral Blood Smear:** basophilic stippling of RBCs
 - Due to aggregates of residual ribosomes
 - **Seen on normal staining**



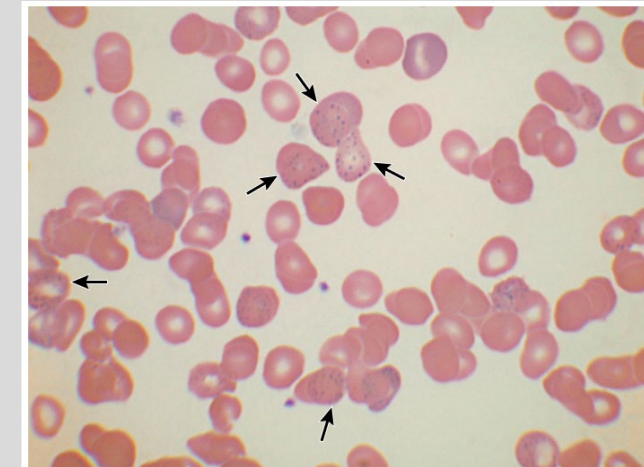
First Aid for the USMLE Step 1 2024,
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Robbins & Cotran Pathologic Basis of Disease, 4e



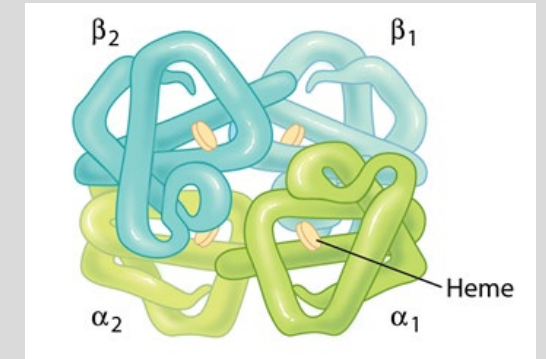
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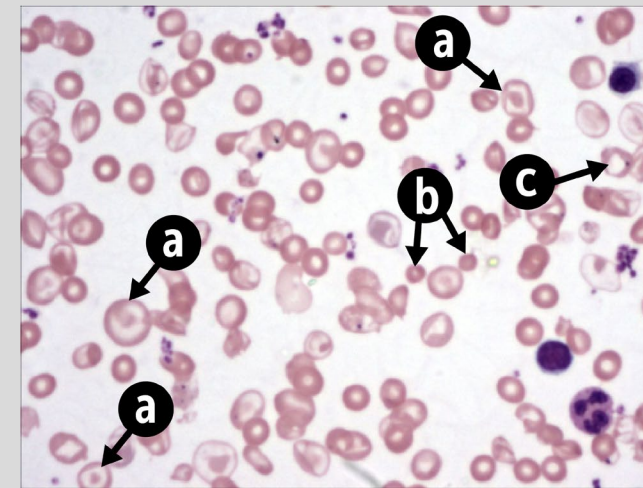
From: Al-Jafar HA, Layton DM, Robertson L, et al. Diagnosis of pyrimidine 5'-nucleotidase deficiency suspected from a blood film. Am J Hematol 2013; 88:1089. <https://onlinelibrary.wiley.com/doi/10.1002/ajh.23555>. Copyright © Wiley Periodicals, LLC. Reproduced with permission of John Wiley & Sons Inc. This image has been provided by or is owned by Wiley. Further permission is needed before it can be downloaded to PowerPoint, printed, shared or emailed. Please contact Wiley's permissions department either via email: permissions@wiley.com or use the RightsLink service by clicking on the 'Request Permission' link accompanying this article on Wiley Online Library (<https://onlinelibrary.wiley.com/>).

Thalassemia - Overview

- Anemia due to decreases synthesis of the globin chains of hemoglobin
 - **Other aspect of hemoglobin is now the issue, heme is okay!**
- **Typically inherited mutations**
 - Evolutionary mutation to help protect against *Plasmodium falciparum* (malaria)
- Separated into α -thalassemia or β -thalassemia based on the decreased production of alpha or beta globin chains
- The body will react to the decrease in one globin chain by increasing production of the other
 - Unbound α or β chains will precipitate within the RBC and cause damage to the cell and decrease cell production and lifespan
 - The membrane is also altered, increasing risk of hemolysis
- **Peripheral Blood Smear: target cells**
 - Nucleated RBCs may be seen in more severe cases
 - More premature cells being sent out
- Lab Findings:
 - **Very low MCV (<70 fL)**
 - **RBC count may be normal in mild thalassemia**



ScholarRx

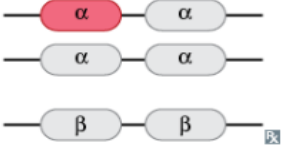
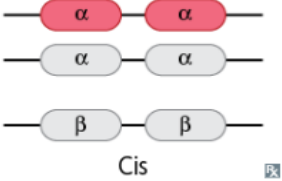
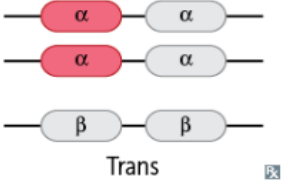
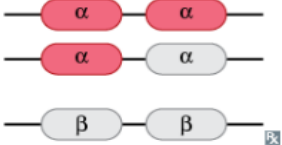
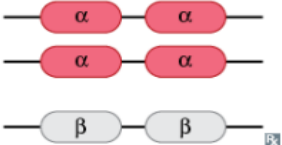


ScholarRx

- a** Target cells
- b** Microcytic cells
- c** Hypochromic cell

Alpha Thalassemia

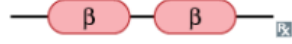
- **α-thalassemia minima:** (1 Gene Deletion)
 - Asymptomatic (silent carrier)
- **α -thalassemia minor:** (2 Gene Deletion)
 - Mild microcytic anemia with **↑ RBC**
 - Cis deletion: occurs on same copy of gene (increased risk of severe anemia in offspring)
 - Deletion occurring on same gene, may transmit no copies of the gene to the offspring
 - Trans deletion: occurs on different copy of gene
 - Cis more common in Asian population
 - Trans-deletion more common in African population
- **Hemoglobin H Disease:** excess β globin form tetramers (β₄)
 - HbH seen on electrophoresis
- **Hemoglobin Barts:** no α-globin chains, γ tetramers form in utero (γ₄)
 - **Incompatible with life (hydrops fetalis)**
 - Hb Barts seen on electrophoresis

# OF α-GLOBIN GENES DELETED ^a	DISEASE	CLINICAL OUTCOME
1 	α-thalassemia minima	No anemia (silent carrier)
2  Cis or  Trans	α-thalassemia minor	Mild microcytic, hypochromic anemia
3 	Hemoglobin H disease (HbH); excess β-globin forms β ₄	Moderate to severe microcytic hypochromic anemia
4 	Hemoglobin Barts disease; no α-globin, excess γ-globin forms γ ₄	Hydrops fetalis; incompatible with life

Beta Thalassemia

- **β -Thalassemia Minor** (β/ β^+): mildest form, mutation on **one** gene
 - Asymptomatic with increased RBC count
 - Due to decreased β -globin, δ -globin will be produced, creating HbA₂ hemoglobin
 - **Electrophoresis:** mild decrease in HbA, increase in HbA₂ (5% vs 2.5% normal) and HbF (2% vs 1% normal)
- **β -Thalassemia Intermedia** (β^+/ β^+ or β^+/ β^0)
 - Variable anemia, depending on production
- **β -Thalassemia Major** (β^0/ β^0): severe, mutation on **both** genes
 - No β -globin chains
 - δ -globin (HbA₂) and γ -globin (HbF) will be produced
 - Electrophoresis: $\uparrow$ HbF, $\uparrow$ HbA₂, no HbA

Point mutation in splice sites or Kozak consensus sequence (promoter) on chromosome 11 $\rightarrow$ $\downarrow$ β -globin synthesis (β^+) or absent β -globin synthesis (β^0). $\uparrow$ prevalence in people of Mediterranean descent.

# OF β -GLOBIN GENES MUTATED+	DISEASE	CLINICAL OUTCOME
1 	β -thalassemia minor	Mild microcytic anemia. $\uparrow$ HbA ₂ .
2 (β^+/ β^+ or β^+/ β^0)	β -thalassemia intermedia	Variable anemia, ranging from mild/ asymptomatic to severe/transfusion-dependent.
2 	β -thalassemia major (Cooley anemia)	Severe microcytic anemia with target cells and $\uparrow$ anisopoikilocytosis requiring blood transfusions ($\uparrow$ risk of 2 ^o hemochromatosis), marrow expansion ("crew cut" on skull x-ray) $\rightarrow$ skeletal deformities, extramedullary hematopoiesis $\rightarrow$ HSM. $\uparrow$ risk of parvovirus B19-induced aplastic crisis. $\uparrow$ HbF and HbA ₂ , becomes symptomatic after 6 months when HbF declines (HbF is protective). Chronic hemolysis $\rightarrow$ pigmented gallstones.
1 (β^+/ HbS or β^0/ HbS)	Sickle cell β -thalassemia	Mild to moderate sickle cell disease depending on whether there is $\downarrow$ (β^+/ HbS) or absent (β^0/ HbS) β -globin synthesis.

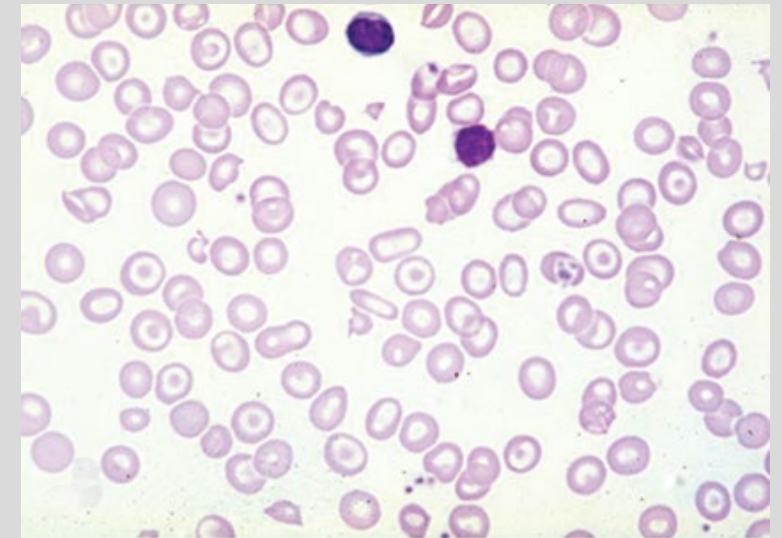
Beta Thalassemia Major

- Most severe form, presents with severe anemia a few months after birth
 - HbF ($\alpha_2\gamma_2$) is temporarily protective (no requirement for β -globin as fetus/infant)
- Unpaired α chains precipitate and damage RBC membranes $\rightarrow$ ineffective erythropoiesis and extravascular hemolysis (removal of RBC by spleen)
- **Erythroid Hyperplasia:**
 - Expansion of hematopoiesis into the skull and facial bones (“crew cut” and “chipmunk facies”)
 - Extramedullary hematopoiesis $\rightarrow$ hepatosplenomegaly
 - Risk of aplastic crisis with parvovirus anemia
- **Peripheral Blood Smear:** microcytic, hypochromic RBCs with target cells and nucleated RBCs
- **Treatment:** chronic RBC transfusions, iron chelation, splenectomy

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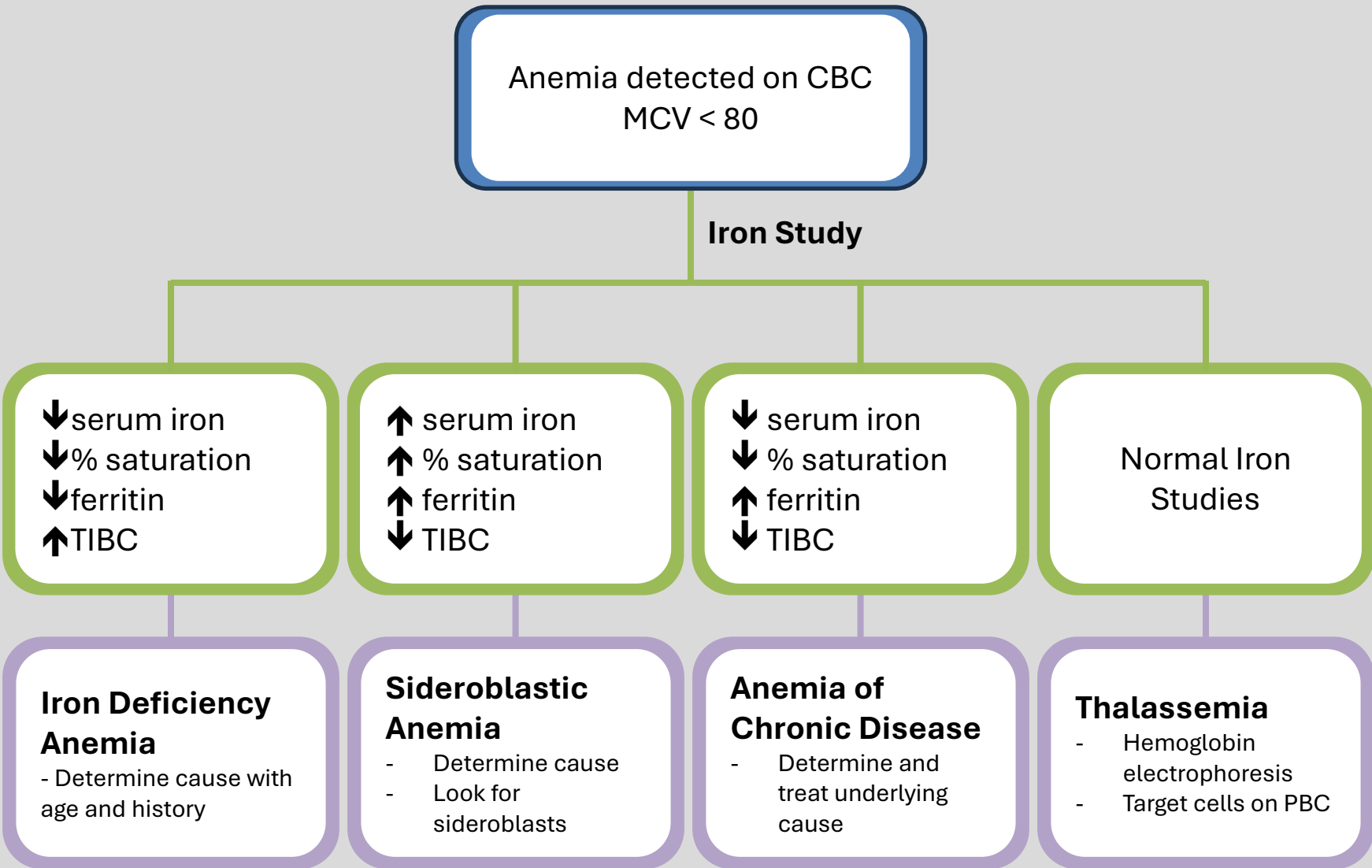


Robbins & Cotran Pathologic Basis of Disease, 4e



Source: Jon C. Aster, H. Franklin Bunn:
Pathophysiology of Blood Disorders, Second Edition

Microcytic Anemia Algorithm



Do.
Make.
Innovate.
Reinvent the Future.

Question #1

A 57-year-old male presents to the ED after being found unresponsive outside of a **liquor store**. According to previous records, he has been admitted 20 times over the last year for similar episodes. He is laying on the hospital bed and IV fluid administration and K⁺ repletion are initiated. His vitals are HR: 98 RR: 19 BP: 108/67 Temp: 99.2 ° F. Labs reveal **increased AST, ALT, GGT**. Hb is 9.8g/dL and MCV is 78fL. What is most likely to be seen in a peripheral blood smear?

- A. Hypersegmented neutrophils
- B. Ringed sideroblasts (we see this on bone marrow aspirate)
- C. Macrocytes
- D. Basophilic stippling**
- E. Target cells

Question #2

A 7-year-old boy is brought to the clinic by his mother after having recently immigrated from Italy. She is concerned because she has a “blood disorder” that runs in her family and wants to make sure her son has not inherited any “bad traits.” The father is healthy and the mother sometimes suffers from mild fatigue and yellowing of the eyes. Physical exam is unremarkable. CBC shows low Hb and low MCV. What type of hemoglobin is most likely to be increased in this boy on electrophoresis?

- A. HbA
- B. HbF
- C. HbA₂
- D. HbS
- E. HbC

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- A. HbA
- B. **HbF**
- C. **HbA₂**
- D. HbS
- E. HbC

Question #3

A 59-year-old female with a 20-year PMHx of rheumatoid arthritis and Hashimoto's thyroiditis presents to the clinic for recent exacerbations of her arthritis as well as a new onset of fatigue. She takes her medications inconsistently because she doesn't believe they work. Physical exam is unremarkable. CBC shows a Hb of 10.1g/dL, an MCV of 78fL and an ESR of 92mm/hr (normal: 0-22mm/hr). You order an iron study. Which lab finding will you most likely find?

- A. ↑ TIBC
- B. ↑ serum iron
- C. ↑ ferritin
- D. ↑ % saturation
- E. ↓ ferritin

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- A.  TIBC
- B.  serum iron
- C.  **ferritin**
- D.  % saturation
- E.  ferritin

Question #4

A 67-year-old male presents to your office because he has been feeling fatigued and has lost 15 pounds in the last 3 months. He is a 30 pack-year smoker and drinks alcohol regularly. He has noticed bloody streaks in his stool. On physical exam you note pale conjunctiva. CBC shows a Hb of 9.2g/dL and an MCV of 78fL. Which type of anemia is he most likely experiencing?

- A. Anemia of chronic disease
- B. Sideroblastic anemia
- C. Iron deficiency anemia
- D. Thalassemia
- E. Lead poisoning

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- A. Anemia of chronic disease
- B. Sideroblastic anemia
- C. **Iron deficiency anemia**
- D. Thalassemia
- E. Lead poisoning

Question #5

An 8-year-old boy who was adopted from China two years ago is brought to the office by his adopted mother for complaints of fatigue, abdominal pain, and bouts of confusion. His teacher has reported similar events. There is no available history on the patient's birth mother. Physical exam shows lines on the gingiva. CBC shows MCV of 72 and iron study show the following:

Serum Iron: 200 (Reference 65-175 $\mu\text{g/dL}$)

Ferritin: 300 (Reference 20-250 ng/mL)

TIBC: 160 (Reference 250-400 $\mu\text{g/dL}$)

% Saturation: 45% (Reference ~33%)

What is the most likely contributing factor to this presentation:

- A. Genetic condition affecting the α -globin chain via an error in the promotor sequence
- B. Genetic condition affecting the β -globin chain via a gene deletion
- C. Nutritional insufficiency
- D. A recent move to a home built in the 1920s
- E. A chronic underlying inflammatory state

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Serum Iron: 200 (Reference 65-175 µg/dL) ↑

Ferritin: 300 (Reference 20-250 ng/mL) ↑

TIBC: 160 (Reference 250-400 µg/dL) ↓

% Saturation: 45% (Reference ~33%) ↑

What is the most likely contributing factor to this presentation:

- A. Genetic condition affecting the α-globin chain via an error in the promotor sequence
- B. Genetic condition affecting the β-globin chain via a gene deletion
- C. Nutritional insufficiency
- D. **A recent move to a home built in the 1920s**
- E. A chronic underlying inflammatory state